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Pd-catalyzed regioselective synthesis of allenyl sulfones *via* sulfinate-enabled isomerization

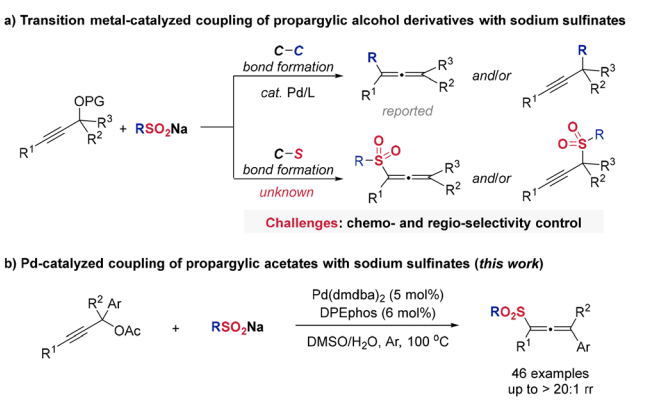
Shiwen Fan,[a] Zhaoqiang Chen,[a] Wenya Li,[a] Hui Qian *[a] and Shengming Ma *[ab]

An efficient Pd-catalyzed C–S coupling strategy for the synthesis of tetrasubstituted allenyl sulfones has been developed. Excess sulfinate uniquely acts as both coupling partner and isomerization promoter, driving the *in situ* generated propargyl sulfone to allenyl sulfone conversion with a high regioselectivity and a broad substrate scope.

Organosulfone compounds, valued for their unique physicochemical properties, are prevalent in natural products and bioactive molecules.[1] Their stability, synthetic accessibility,[2] and diverse reactivity enable participation in key transformations such as Julia–Kocienski olefination,[3] Claisen rearrangement,[4] and Truce–Smiles rearrangement.[5] Furthermore, they undergo desulfonylative conversions *via* transition metal catalysis, organocatalysis, photocatalysis, or electrocatalysis.[6] Consequently, the synthesis of sulfone compounds has attracted significant interest, driving the development of numerous methods.[7] Allenyl sulfones, a distinct subclass that retains sulfonyl functionality while incorporating allene groups, are versatile building blocks in organic synthesis.[8] Therefore, the development of efficient synthetic methods to access allenyl sulfones has attracted considerable scientific interest.[9]

In prior work, our group established a methodology for the synthesis of allenes *via* SO_2 extrusion in the transition-metal-catalyzed reaction of propargylic alcohol derivatives with sodium sulfinates (Scheme 1a).[10] Nevertheless, the synthesis of allenyl sulfones poses significant challenges in both chemoselectivity and regioselectivity. In this context, we report a Pd-catalyzed C–S coupling between propargylic acetates and sodium sulfinates. This reaction affords tetrasubstituted allenyl sulfones with an excellent regioselectivity and a broad functional group tolerance under mild conditions.

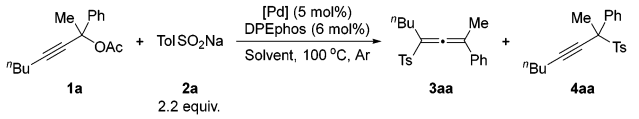
Our study initiated with the reaction of propargylic acetate **1a** and sodium *p*-toluenesulfinate **2a**. Employing $Pd_2(dba)_3 \cdot CHCl_3$ as catalyst in a DMF/H_2O mixed solvent at 100 °C for 16 h, the reaction afforded the product **3aa** in 76% yield (Table 1, entry 1. For detailed ligand screening, see supplementary information (SI) Table S1). Systematic optimization of the reaction conditions was focused on solvent, reaction time, and palladium source. The reactions in other mixed solvents of DMA or dioxane with water led to reduced yields (Table 1, entries 2 and 3). However, the reaction in DMSO/H_2O afforded the **3aa** in an improved yield of 79% (Table 1, entry 4). Reducing the reaction time to 1 h achieved a complete conversion with an 81% yield (Table 1, entry 5). While other Pd sources afforded comparable results (Table 1, entries 6 and 7), utilizing $Pd_2(dba)_3$ significantly enhanced the yield to 91% (Table 1, entry 8). Control experiments confirmed the essential role of palladium, as no reaction occurred in its absence (Table 1, entry 9). Omission of the ligand or water resulted in diminished yields (64% and 53%, respectively) and the detection of by-product propargylic sulfone **4aa** (Table 1, entries 10 and 11). Upon running the reaction on 1 mmol, an 80% yield of **3aa** was obtained (Table 1, entry 12). Notably, replacement of $Pd_2(dba)_3$ with $Pd(dmdba)_2$ under the otherwise same conditions further improved the yield to 87% (Table 1, entry 13). Consequently, the optimized conditions for further study were defined as follows: **1a** (1.0 mmol), **2a** (2.2 equiv.),



Scheme 1 Previous work and our strategy for the synthesis of allenyl sulfones.

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Table 1 Optimization of the reaction conditions[a]

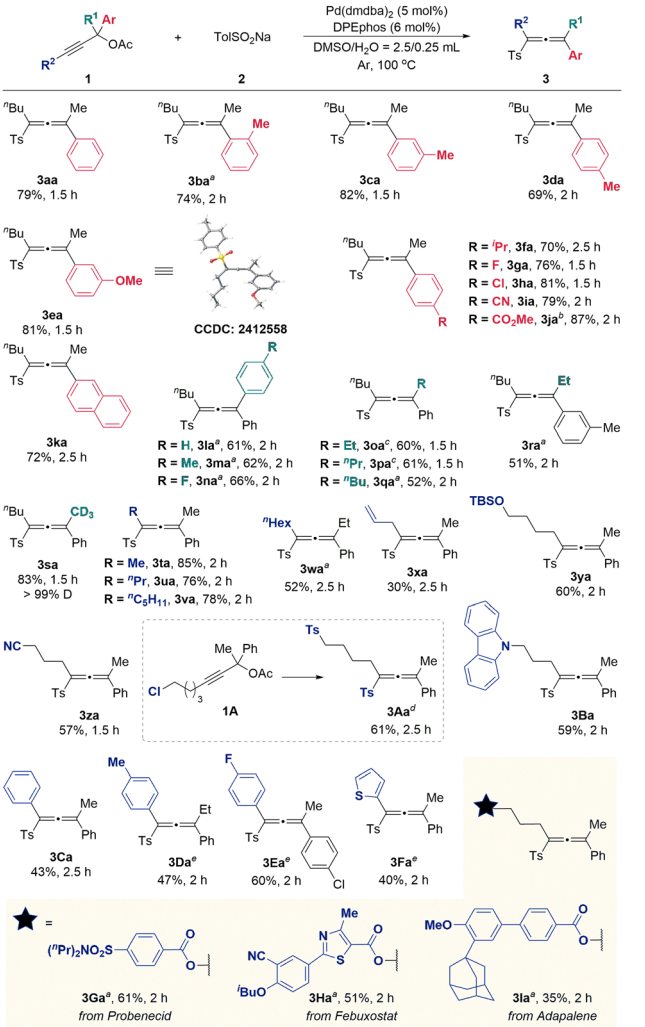


Entry	[Pd]	t	Solvent	Yield 3aa	Yield 4aa
1	$Pd_2(dba)_3{\cdot}CHCl_3$	16 h	DMF/H_2O = 1.0/0.1 mL	76%	—
2	$Pd_2(dba)_3{\cdot}CHCl_3$	16 h	DMA/H_2O = 1.0/0.1 mL	63%	Trace
3	$Pd_2(dba)_3{\cdot}CHCl_3$	16 h	Dioxane/H_2O = 1.0/0.1 mL	47%	14%
4	$Pd_2(dba)_3{\cdot}CHCl_3$	16 h	DMSO/H_2O = 1.0/0.1 mL	79%	—
5	$Pd_2(dba)_3{\cdot}CHCl_3$	1 h	DMSO/H_2O = 1.0/0.1 mL	81%	—
6	$Pd(OAc)_2$	1 h	DMSO/H_2O = 1.0/0.1 mL	82%	—
7	$Pd(dmdba)_2$	1 h	DMSO/H_2O = 1.0/0.1 mL	84%	—
8	$Pd_2(dba)_3$	1 h	DMSO/H_2O = 1.0/0.1 mL	91%	—
9[b]	—	1 h	DMSO/H_2O = 1.0/0.1 mL	—	—
10[b]	$Pd_2(dba)_3$	1 h	DMSO/H_2O = 1.0/0.1 mL	64%	8%
11	$Pd_2(dba)_3$	1 h	DMSO = 1.0 mL	53%	22%
12[c]	$Pd_2(dba)_3$	2 h	DMSO/H_2O = 2.5/0.25 mL	80%	—
13[c]	$Pd(dmdba)_2$	2 h	DMSO/H_2O = 2.5/0.25 mL	87%	—

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (2.2 equiv.), [Pd] (5 mol%), DPEphos (6 mol%) in solvent/H_2O (1.0/0.1 mL) at 100 °C under Ar atmosphere. The yield of **3aa** and **4aa** was determined by ^1H NMR analysis using CH_2Br_2 as the internal standard. [b] Without DPEphos. [c] Reaction was conducted using **1a** (1.0 mmol) in DMSO/H_2O (2.5/0.25 mL) at 100 °C under Ar atmosphere. DPEphos is bis(diphenylphosphanylphenyl)ether; dba is dibenzylideneacetone; dmdba is 3,5,3′,5′-dimethoxydibenzylideneacetone.

$Pd(dmdba)_2$ (5 mol%), DPEphos (6 mol%), DMSO/H_2O (2.5 mL/0.25 mL), 100 °C, Ar atmosphere, 2 h (Table 1, entry 13).

With the optimized reaction conditions, we evaluated the substrate scope using diverse propargylic acetates and sodium sulfinates (Scheme 2). The substrates with either electron-donating (**1b–1f**) or electron-withdrawing (**1g–1j**) groups on the aryl rings (Ar) reacted smoothly with *p*-toluenesulfinate **2a**, affording the corresponding allenyl sulfones **3ba–3ja** smoothly. Single-crystal X-ray diffraction of **3ea** unambiguously confirmed the product structures. Notably, steric effects exerted minimal influence: *ortho*-methyl (**1b**), *meta*-methyl (**1c**), and *para*-methyl (**1d**) substrates all delivered products **3ba–3da** with comparable yields. Similarly, 2-(naphth-2-yl)oct-3-yn-2-yl acetate (**1k**) provided **3ka** in 72% yield. Substitution patterns at R^1 were also explored: aryl-substituted substrates **1l–1n** generated allenyl sulfones **3la-3n** in moderate yields, while the substrates with alkyl chains of varying lengths (**1o–1r**) consistently produced the corresponding allenes **3oa-3ra**. The deuteromethylated analogue **1s** furnished deuterated product **3sa** in an excellent yield. Then, we examined the R^2 substituent: a series of substrates containing reactive functional groups and varying carbon chain lengths reacted smoothly, yielding moderate to good results (**3ta-3Ba**). When propargylic acetate **1A** bearing a terminal chloro group was employed, the disulfone product **3Aa** was obtained *via* Pd-catalyzed sulfonylation and nucleophilic substitution. Aryl R^2 groups (**1C–1F**) were also tolerated (**3Ca–3Fa**). Finally, pharmaceutically relevant fragments from probenecid, febuxostat, adapalene (**1G–1I**) were successfully incorporated, delivering allenes **3Ga–3Ia** in moderate yields. The present reaction conditions show limitations in preparation of di- and trisubstituted allenyl sulfones (see Scheme S1 in SI for details).

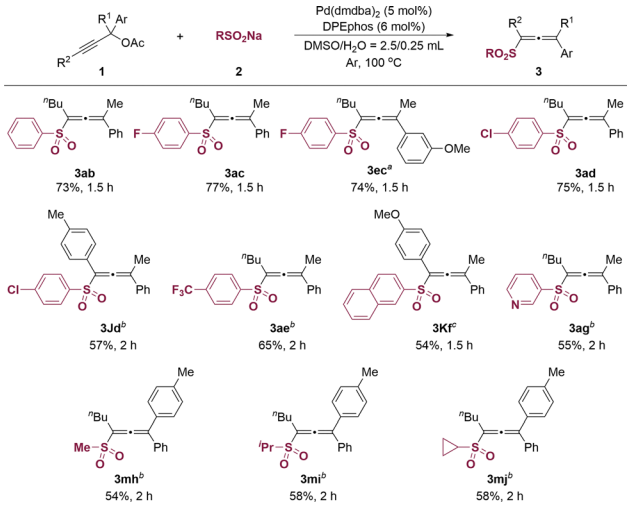


Scheme 2 Substrate scope of propargylic acetates. Reaction conditions: **1** (1.0 mmol), **2** (2.2 equiv.), $Pd(dmdba)_2$ (5 mol%), DPEphos (6 mol%) in DMSO/H_2O (2.5/0.25 mL) at 100 °C under Ar atmosphere. [a]$Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%) was used. [b]**1** (2.0 mmol) was used. [c]$Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%), DMSO/H_2O (2.5/0.5 mL) was used. [d]**2** (4.4 equiv.), $Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%) was used. [e]**1** (2.0 mmol), $Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%) was used.

We next examined the scope of sodium sulfinates (Scheme 3). Aryl sodium sulfinates reacted smoothly with propargylic acetates, affording allenyl sulfones **3ab–3ae** in moderate to good yields. Critically, all substrates delivered comparable results as compared to *p*-toluenesulfinate **2a**, indicating minimal electronic influence. 2-Naphthalenesulfonyl and pyridine-3-sulfonyl variants also participated efficiently, yielding **3Kf** and **3ag** in moderate yields. The alkyl sodium sulfinates, including methyl, isopropyl and cyclopropyl, also demonstrate decent reactivity (**3mh–3mj**).

To highlight the synthetic utility, a gram-scale reaction delivered 1.0321 g of **3ya** (55% yield, Scheme 4). Under the Cu catalysis, **3ya** underwent C–S bond functionalization with Grignard reagents to construct a C–C bond.[11] Subsequent desilylation with TBAF afforded allenol **4** in 50% yield over two steps. Alternatively, direct desilylation of **3ya** efficiently provided allenol **5** with a yield of

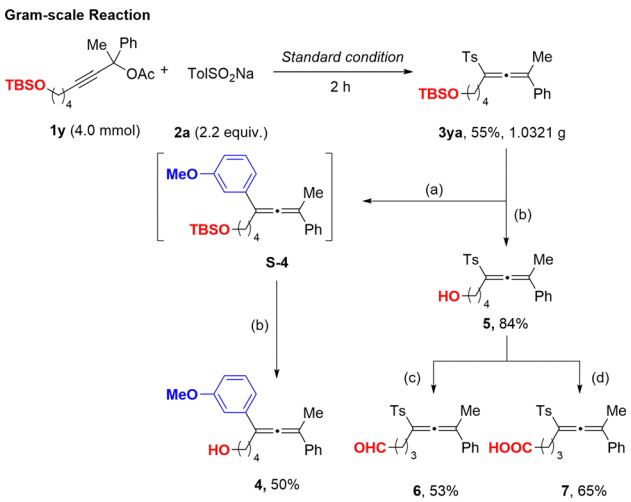
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Scheme 3 Substrate scope of sodium sulfinates. Reaction conditions: **1** (1.0 mmol), **2** (2.2 equiv.), $Pd(dmdba)_2$ (5 mol%), DPEphos (6 mol%) in DMSO/H_2O (2.5/0.25 mL) at 100 °C under Ar atmosphere. [a]**1** (2.0 mmol) was used. [b]$Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%) was used. [c]**1** (2.0 mmol), $Pd(dmdba)_2$ (7.5 mol%), DPEphos (9 mol%) was used.

84%. Selective oxidation of allenol **5** using $Fe(NO_3)_3{\cdot}9H_2O$/TEMPO with NaCl or KCl enabled divergent synthesis of aldehyde **6** in 53% yield and carboxylic acid **7** in 65% yield.[12]

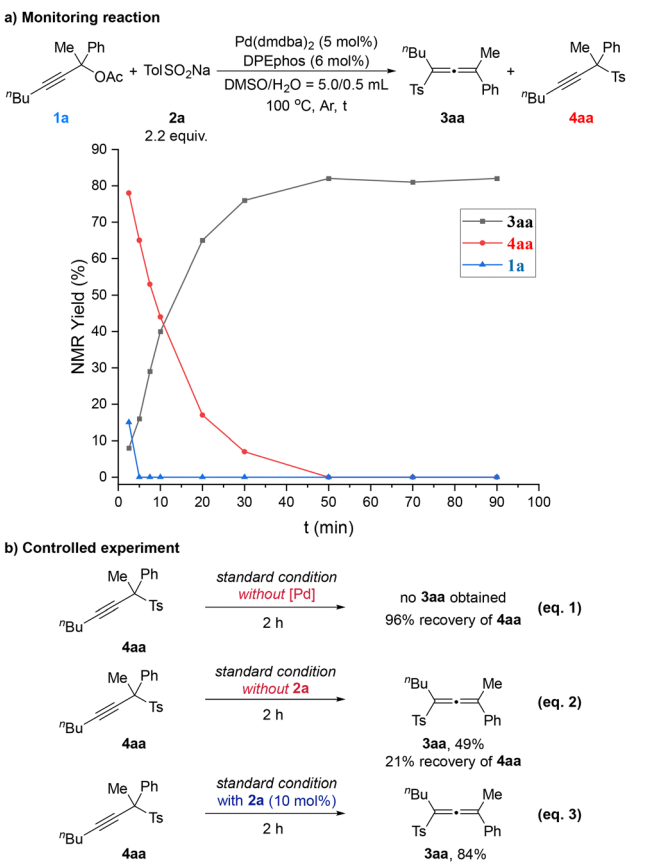
Mechanistic investigation commenced with the monitoring of the reaction of propargylic acetate **1a** and sodium *p*-toluenesulfinate **2a** under the standard conditions (Scheme 5a). Within 2.5 minutes, **1a** underwent 85% conversion, yielding propargylic sulfone **4aa** as the primary intermediate (78%) alongside minimal allenyl sulfone **3aa** (8%). Complete substrate depletion at 5 minutes afforded alkyne **4aa** (65%) and allene **3aa** (16%), confirming



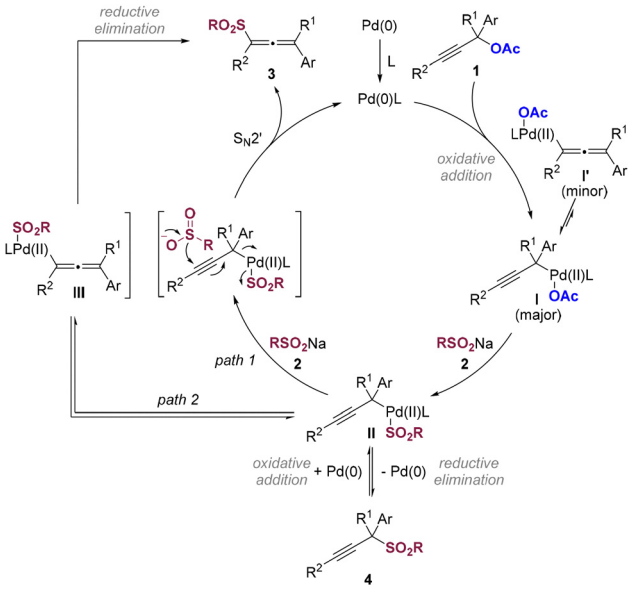
Scheme 4 Gram-scale reaction and synthetic applications. Reaction conditions for synthetic applications: [a]**3ya** (0.2 mmol), $Cu(acac)_2$ (10 mol%), 3-OMe-C_6H_4MgBr (1.8 equiv.), THF (0.4 mL), rt, 24 h. [b]**3ya** or **S-4** (0.2 mmol), TBAF (5.0 equiv.), THF (4.0 mL), rt, 2 h. [c]**5** (0.2 mmol), $Fe(NO_3)_3{\cdot}9H_2O$ (15 mol%), TEMPO (15 mol%), NaCl (15 mol%), DCE (2.0 mL), rt, 2 h. [d]**5** (0.2 mmol), $Fe(NO_3)_3{\cdot}9H_2O$ (25 mol%), TEMPO (25 mol%), KCl (25 mol%), DCE (2.0 mL), rt, 25 h. TEMPO is 2,2,6,6-tetramethylpiperidinooxy; DCE is 1,2-dichloroethane.

concurrent isomerization of alkyne **4aa** to allene **3aa**. This transformation progressed continuously until exclusive formation of **3aa** at 50 minutes with 82% yield. Control experiments established the essential role of palladium: **4aa** remained inert without the Pd catalyst (eqn (1) in Scheme 5b). Omission of **2a** permitted only partial isomerization to **3aa** (Scheme 5b, eqn (2)), whereas inclusion of Pd and **2a** drove complete conversion to **3aa** even with 10 mol% of sulfinate **2a** (Scheme 5b, eqn (3)), demonstrating its dual function as a reactant and an isomerization promoter.

Based on mechanistic studies, we have proposed a catalytic cycle (Scheme 6): propargylic acetate **1** undergoes oxidative addition with Pd(0), predominantly forming propargyl-palladium(II) complex **I** alongside a minor amount of allenyl-palladium(II) isomer **I′**.[13] In the dominant pathway, subsequent ligand exchanging with sodium sulfinate **2** affords intermediate **II**, which undergoes reductive elimination to deliver propargylic sulfone **4** while regenerating Pd(0). Intermediate alkyne product **4** then re-enters the catalytic cycle *via* oxidative addition to regenerate propargylic palladium complex **II**, and there are two possible pathways for the conversion of intermediate **II** to allenyl sulfone **3**: Path 1 involves concerted S_N2' displacement by sulfinate **2**, directly yielding **3** and releasing Pd(0); Path 2 proceeds *via* direct isomerization of **II** to allenyl-palladium(II) complex **III**, followed by reductive elimination to furnish **3** and Pd(0), thereby completing the catalytic cycle. Direct conversion *via* **I′** cannot be excluded



Scheme 5 Mechanistic studies.



Scheme 6 The proposed mechanism.

as supported by the fact that a trace amount of **3** was formed at 2.5 minutes (Scheme 5a).

In summary, we have established an efficient Pd-catalyzed C–S bond formation strategy for synthesizing tetrasubstituted allenyl sulfones. Critical to this methodology is the use of excess sodium sulfinate, which drives the isomerization of propargyl sulfone intermediates to the target allenyl sulfones. This approach delivers an exceptional regioselectivity and a broad substrate scope, encompassing diverse functional groups and pharmaceutical motifs. Our laboratory is currently conducting further research on this topic.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information is available. See DOI: https://doi.org/10.1039/d6cc02404a.

CCDC 2412558 (**3ea**) contains the supplementary crystallographic data for this paper.[14]

Acknowledgements

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